# Two symmetries, one task: what three LLM ensembles named, built, and inferred

Banana Labs · ShinkaEvolve on transfer-sn, three arms × 50 generations, 144 LLM proposals

## What the proposer was told

The hidden truth is that each record is the 28 upper-triangle adjacency bits of an 8-vertex graph and the label is connectedness, so relabelling vertices permutes qubits and features jointly and the label never changes. The task is invariant under the symmetric group $S_8$. None of that is stated. This is the complete task message, verbatim:

> You are an expert in variational quantum circuit design.
>
> Goal: Evolve the ANSATZ_SPEC for an 8-qubit classifier of binary-feature records. The fixed code will train each candidate with Adam in a simulated PennyLane circuit.
>
> Fixed architecture: 8 qubits. Each record has 28 binary features; the fixed feature map encodes feature k as a two-qubit phase coupling (IsingZZ) on a fixed qubit pair given by the dataset's pair table. Data re-uploading l=3; the candidate ansatz block is applied p=2 times after each upload, with independent parameter copies per block. Readout: the mean of all single-qubit Z expectations, passed through a trainable gain and bias, regressed to the +1/-1 label.
>
> Only edit the EVOLVE-BLOCK, especially ANSATZ_SPEC. Do not change the fixed training loop, dataset handling, measurements, l, p, or feature map.
>
> Formal ANSATZ_SPEC schema: Single-qubit parametrized gates: {"gate": "RX"|"RY"|"RZ", "wire": int 0..7, "param": "name"}. Fixed two-qubit gates: {"gate": "CNOT"|"CZ", "wires": [first, second]}. Parametrized controlled rotations: {"gate": "CRX"|"CRY"|"CRZ", "wires": [control, target], "param": "name"}
>
> Connectivity: two-qubit gates may act on ANY pair of distinct qubits.
>
> Parameter sharing: Reusing the same param string shares that parameter within one ansatz block.
>
> Starting point: The seed block applies independent RY and RZ rotations on every qubit, a line of CZ gates, and a final RZ layer.
>
> Candidate quality: Improve validation/test accuracy, reduce the train-test gap, reduce L2 loss, use parameters efficiently, and reach high accuracy in fewer Adam steps.
>
> Invalid candidates are rejected for unsupported gates, bad wires, non-finite metrics, or too many parameters.

Two leaks matter. Features are said to map onto qubit pairs, and $28 = \binom{8}{2}$, so a model that counts can deduce the features are all unordered pairs of the 8 qubits. The sharing mechanism is handed over too. What is never supplied is which gates should share, or why. The word symmetry never appears.

## Two vocabularies, measured separately

Counting the word "symmetry" conflates two claims, so we count two regexes over the proposer's own patch name and description, never over evaluator feedback. General covers any symmetry idea (symmetr*, invarian*, mirror*, plus the permutation terms). Permutation-specific covers only vocabulary presupposing the right concept: equivarian*, permut*, orbit*, relabel*, exchangeab*, S_8. The distinction is not pedantic: the mid arm's "invariant" is "the final RZ layer is effectively measurement-invariant for the Z readout", which concerns the measurement basis, not the data.

## Findings

1. The arms named different symmetries. Mid named a symmetry in 81% of proposals and permutation symmetry in none; three quarters of its notes are about mirror symmetry of the circuit layout (mirror_butterfly_crz, "4 params shared across mirror-symmetric qubit pairs"). Frontier is the only arm that ever used permutation vocabulary, in 31% of proposals. Weak named almost nothing, 8.5%.

2. Both arms built the symmetry they named. Mid's follow-through on mirror symmetry is 97%: 35 of its 36 mirror claims pair wire $i$ with wire $7-i$ under a shared angle. Frontier's follow-through on permutation symmetry is 100%: all 15 permutation claims contain a family of one angle tied across all 8 wires. Neither arm was confabulating. They differed in which symmetry they hypothesised, and mid's is not a symmetry of this task, so building it faithfully bought nothing: mid finished at 0.273 against frontier's 1.100.

3. The discovery was proposed, not inherited. Frontier introduced the tied orbit four times, first at generation 3. Each time the parent lacked the structure, no inspiration program shown to the model contained it, and the first two notes cite no prior program. Generation 3 is a full rewrite cutting 24 free angles to 4 without using the word symmetry; generation 5 is a diff to 3 angles that does. The structure came first, the vocabulary second. Weak introduced the orbit nine times and retained it four, rediscovering and losing it. Mid never introduced it at all.

| | weak | mid | frontier |
|---|---|---|---|
| reasoning effort | low | medium | xhigh |
| named any symmetry | 8.5% | 81.2% | 40.8% |
| named permutation symmetry | 0% | 0% | 30.6% |
| named mirror symmetry | 2.1% | 75.0% | 2.1% |
| built 8-wire tied orbit | 27.7% | 0% | 83.3% |
| orbit introductions / retained | 9 / 4 | 0 / 0 | 4 / 36 |
| first orbit built | gen 14 | never | gen 3 |
| best score | 0.834 | 0.273 | 1.100 |

Weak is the instructive case. It reached 0.834, above the hand-built $S_8$-equivariant baseline at 0.675, having never once named the symmetry. Its best program splits the wires into two half-orbits of four, invariant under $S_4 \times S_4$ rather than $S_8$, reached by mechanical parameter reduction. Naming the symmetry is sufficient but not necessary; building it is what pays.


A: share of each arm's proposals whose patch note uses that vocabulary. B: solid = named the symmetry, hatched = built the matching structure (8-wire tied family for permutation; shared $i \leftrightarrow 7-i$ angle for mirror). C: squares mark generations whose block contains an 8-wire tied orbit, triangles mark introductions where the parent lacked it. $n = 47/48/48$ proposals with a parseable ANSATZ_SPEC. Vocabulary counts detect wording, not understanding; the structural ones do not.